

The assessment of baroreflex sensitivity in patients with chronic kidney disease: implications for vasomotor instability

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Purpose of review

Autonomic dysfunction is well recognized in chronic kidney disease. The baroreflex arc is an important component of the autonomic nervous system and influences vasodilatation and heart rate in response to information derived from baroreceptors located in the aorta and common carotid arteries. Appropriate regulation of systemic blood pressure is therefore dependent upon the integrity and normal function of the baroreflex arc. Vasomotor instability during haemodialysis has traditionally been identified as a pathophysiological state largely due to the failure of compensatory mechanisms during ultrafiltration. This review article discusses autonomic dysfunction as a key factor contributing to the systemic haemodynamic instability that may occur during dialysis.

Recent findings

Relationships have begun to be established between markers of reduced baroreflex sensitivity and abnormal cardiovascular structure, such as left ventricular dysfunction, left ventricular hypertrophy and arterial stiffness.

Summary

Pathological relationships between cardiovascular structure and function remain important in the understanding of both haemodynamic instability and cardiovascular mortality in chronic kidney disease patients. Understanding the associations between conventional markers of haemodynamic instability and autonomic function will allow early identification of patients likely to benefit from individualizing dialysis therapy.

Keywords

autonomic neuropathy, baroreflex sensitivity, continuous ambulatory peritoneal dialysis, haemodialysis, haemodynamic instability, heart rate variability, intra-dialytic hypotension

Abbreviations

BP	blood pressure
BRS	baroreflex sensitivity
CAPD	continuous ambulatory peritoneal dialysis
CKD	chronic kidney disease
HRV	heart rate variability
IDH	intra-dialytic hypotension

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Introduction

Cardiovascular disease is the leading cause of mortality in patients with chronic kidney disease (CKD) [1]. Indeed the incidence of cardiac failure and renal failure is increasing and the combined effects contribute to an overall bleak prognosis [2,3]. It has been highlighted that pathological relationships between cardiac output, relative blood volume and blood pressure (BP) undoubtedly contribute to the overall mortality but not all aspects of the profound haemodynamic changes, both structural and functional, can be explained on this basis alone. Other factors such as increased renin–angiotensin activation, oxidative stress, structural myocardial changes, inflammation and increased sympathetic activity also appear to contribute [4•]. Short-term regulation of BP is largely controlled by appropriate autonomic nervous activity through the baroreflex arc. Baroreflex arc activity, also known as baroreflex sensitivity (BRS), is well recognized as a composite marker of the overall integrity of the autonomic nervous system [5]. Intact autonomic function is therefore essential in the maintenance of normal cardiovascular function and treatment tolerability during dialysis, yet impaired BRS has been widely documented in CKD [6–8]. Vita and colleagues [9] investigated both the parasympathetic and sympathetic responses to clinical autonomic tests and heart rate variability (HRV) in 30 patients receiving chronic haemodialysis. Of the patients 53% had parasympathetic dysfunction, isolated in 40% and combined with sympathetic dysfunction in the remainder, and 37% had borderline parasympathetic abnormalities. Interestingly, they did not identify isolated sympathetic dysfunction. More recently, the same group have confirmed these results, but concluded that the previous statements, that the major aberration of autonomic dysfunction in CKD involved the parasympathetic pathway, would have to be modified to include a more widespread pattern of abnormality [10•]. Measurement of resting (or spontaneous) BRS and investigating

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the ability to alter BRS in changing haemodynamic environments may also be important. BRS is not a fixed entity and a variety of interventions such as exercise, drugs and the impact of electrolytes on the autonomic nervous system have all been reported [11–15].

This article will concentrate upon the contribution of both baroreceptor dysfunction and impaired baroreflex action to systemic haemodynamic instability.

Baroreflex sensitivity

Autonomic function is essential in the overall regulation of systemic BP. The baroreflex arc ensures that changes in BP result in activation of compensatory mechanisms thereby restoring BP to normal limits. For instance, a rise in BP results in an increase in trans-mural stretch within the aorta and the common carotid arteries (Fig. 1). This causes activation of the baroreceptors located within the arterial wall. Increased firing from these receptors is transmitted via the afferent nerves (vagus and glossopharyngeal) to the nucleus tractus solitarius within the midbrain. Sympathetic and parasympathetic pathways influence vessel tone and heart rate thereby constituting the efferent response. A rise in BP ultimately results in an increase in the R–R interval and vasodilatation, and hence a restoration of the previous level of BP. BRS is therefore determined by the mechanical properties of the arterial wall and the parasympathetic and sympathetic nervous systems.

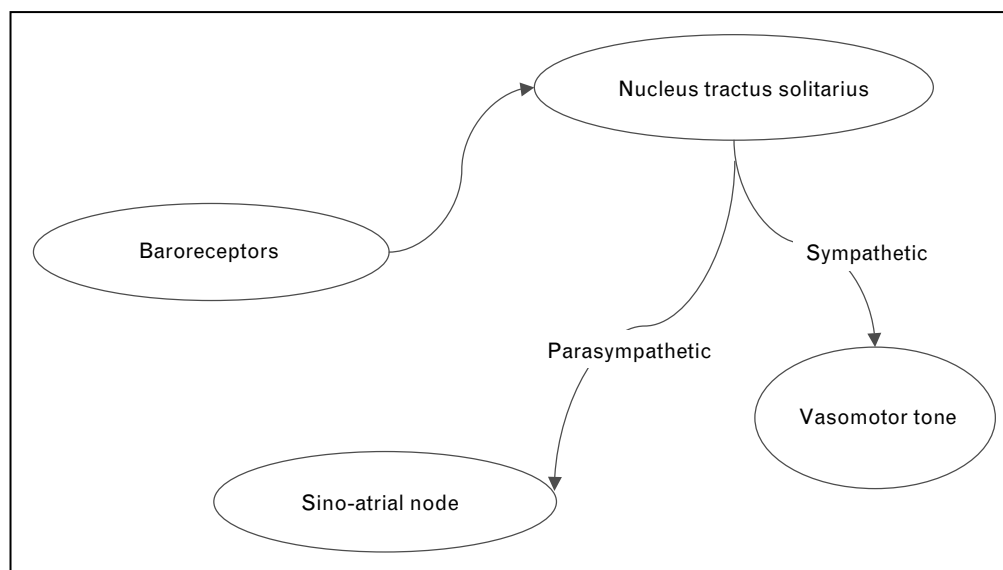
Assessment of baroreflex sensitivity

Initially, measurements of BRS required invasive, intra-arterial methods to assess heart rate responses to pressors (phenylephrine) or vasodilators (nitroprusside) [16,17].

Consequently, the regression of pulse interval against systolic BP represents an index of BRS [18]. Investigation of autonomic function has largely been limited to resting BRS, due to the practical and logistical difficulties of obtaining BRS measurements in a dynamic state. Clinical tests investigating the parasympathetic system involve measuring the heart rate response to respiration, orthostatic change and the Valsalva manoeuvre. Sympathetic integrity can be assessed by evaluating the BP response to both orthostatic change and to pressor stimuli such as handgrip and cutaneous cold. Robinson and Carr [5] published a thorough review in 2002 of the literature pertaining to the investigation of autonomic function in patients with dialysis-independent CKD, and patients receiving chronic haemodialysis or continuous ambulatory peritoneal dialysis (CAPD) [5].

Novel tools for monitoring haemodynamic responses and BRS are now available. Non-invasive pulse-wave analysis permits continuous measurement of beat-to-beat BP. The Finometer (TNO Instruments, Amsterdam, The Netherlands), and its predecessor the Finapres (TNO Instruments), utilize photoplethysmography [19] to derive a pulse wave from a finger pressure cuff containing an infrared optical source and detection unit. The pulse wave is then subject to mathematical analysis based on the transport characteristics of peripheral blood vessels to reconstruct a central pressure waveform. This in turn is analysed to calculate a wide variety of haemodynamic variables including BP, heart rate, BRS, cardiac output, peripheral resistance, the Buckberg index and the ejection duration on a beat-to-beat basis [20,21]. Since spontaneous BRS may be calculated from the regression of pulse interval on systolic BP, haemodynamic data can be

Figure 1. The baroreflex arc



collected and stored for off-line analysis of BRS. This has advantages if collecting data in a physiologically changing state, such as haemodialysis.

HRV is the conventionally accepted term to describe variations in both instantaneous heart rate and the R–R interval [22]. HRV has been shown to be a strong, independent predictor of mortality following myocardial infarction [23,24], indicating that cardiac autonomic control is of critical importance in the overall maintenance of cardiovascular stability. Power spectral analysis can evaluate changes in beat-to-beat cardiovascular control by analysing fluctuations in heart rate. Evaluation of HRV can be through either time domain (useful for evaluating long periods of monitoring), or frequency domain measures. Both methods provide basic information of power (variance) distribution as a function of frequency. Spectral components from frequency domain measures can be separated into areas of very low frequency, low frequency and high frequency. In essence, vagal (parasympathetic) activity is the major contributor to the high frequency component and the low/high frequency ratio is considered by some to represent the sympathovagal balance [22]. Representation of low and high frequency in normalized units emphasizes the controlled and balanced behaviour of the two branches of the autonomic nervous system. It is accepted, however, that this separation should be interpreted with caution [25]. Further detailed discussion of the measurement of individual components of the baroreflex arc, the parasympathetic and sympathetic pathways, is beyond the scope of this article.

Intra-dialytic hypotension and autonomic function

Intra-dialytic hypotension (IDH) occurs in approximately 30% of patients [26], even in those broadly considered 'stable' on dialysis. It consumes significant resources both in medical and nursing time, and may necessitate shortening dialysis times. More significantly, it has recently been associated with increased mortality [27,28^{••}]. Multiple mechanisms have been implicated although there is a paucity of data as to the individual, relative contributions to overall mortality. Shoji *et al.* [28^{••}] discuss the possibility of haemodynamically mediated cerebral ischaemia, age, atherosclerosis, poor cardiac reserve, large volumes of ultrafiltration and diabetes as all contributing to dialysis-associated hypotension. Interestingly, each of these areas is associated, either directly or indirectly, with reduced BRS [10,29–32]. Passauer and colleagues [33] suggest that there are two main groups of patients that are susceptible to dialysis hypotension. Firstly, there are those that are normotensive or hypertensive who are prone to intra-dialytic decreases in systolic BP of 30 mmHg resulting in symptomatic hypotension. The second group includes those that are chronically hypotensive during haemodialysis and the

inter-dialytic interval. Sensitive relationships between ultrafiltration, refill and various compensatory mechanisms exist to maintain cardiovascular stability during haemodialysis [34]. Four pathophysiological conditions have been investigated to elucidate the abnormalities that result in cardiovascular instability during haemodialysis: insufficient refill; autonomic dysfunction; left ventricular diastolic dysfunction; and imbalance of vasoactive agents. Ultrafiltration activates the cardiopulmonary reflex arcs of the autonomic nervous system, resulting in blood redistribution from skin and splanchnic circulations to central compartments [35,36]. Furthermore, increasing sympathetic activity increases both heart rate and systemic vascular resistance thereby restoring cardiovascular stability. Normal functioning of the baroreflex arc is therefore essential in the maintenance of BP during the physiological challenge of dialysis. In addition, another autonomic mechanism, the Bezold–Jarisch reflex, is activated by a severe reduction in venous return. To prevent myocardial ischaemia, this reflex results in a phase of sympathetic activity, followed by a reduction in myocardial workload through a period of bradycardia and reduced afterload [37]. Barnas and colleagues [38] investigated HRV during sessions of haemodialysis that included periods of IDH in eight hypotension-prone patients, compared with 11 patients considered to be cardiovascularly stable. They discovered that in those that maintained BP, an increase in heart rate was accompanied by an increase in the low frequency component, indicating intact sympathetic activity. Conversely, in those that developed severe symptomatic hypotension, decreased sympathetic and increased parasympathetic activity was detected, resulting in bradycardia. They concluded that the Bezold–Jarisch reflex was being activated due to a reduction in intravascular volume and cardiac filling and that this was the cause of sudden IDH. Intact autonomic function is certainly crucial in maintaining haemodynamic stability but the Bezold–Jarisch reflex may not be the only significant factor accompanying IDH. Rubinger and colleagues [39^{••}] usefully contributed to this area with a study investigating HRV in 56 chronic haemodialysis patients either stable or prone to IDH. Patients were investigated during a 24-h period thereby incorporating fluctuations in HRV during haemodialysis, day and night. Sympathovagal balance was also found to be lower in those prone to instability. They concluded that a significantly reduced HRV was associated with haemodynamic instability, although this relationship did not remain apparent if the sympathetic and parasympathetic components were separated into constituent pathways.

Relationships between cardiac structure and markers of autonomic activity

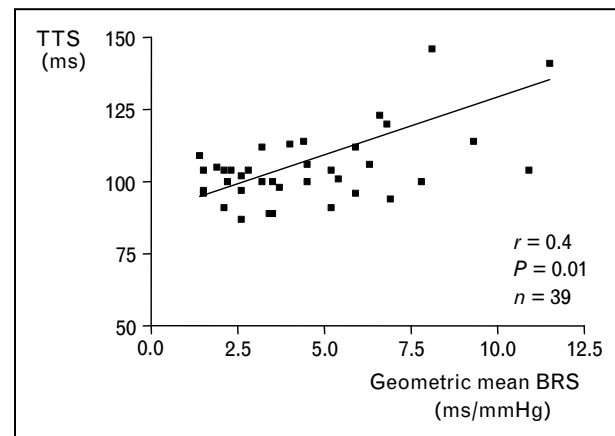
Mathematical models provide extra information on the role of HRV in the prediction of altered cardiovascular

control in those with CKD. Lerma and colleagues [40] developed a mathematical model and observed increased peripheral resistance, reduced BRS and reduced stroke volume in CKD patients. They concluded that the cardiac pump response to haemodynamic instability was restricted. Although measures of BRS or HRV were not included in a study by Boon *et al.* [41], these investigators found that those that became hypotensive in response to blood volume reduction as opposed to those that remained haemodynamically stable differed in their adaptive stroke volume response. It remains to be proven if stroke volume is another area that is intrinsically related to BRS. Karayaylali *et al.* [42] studied left ventricular function and HRV in order to investigate a relationship between cardiac function and autonomic neuropathy. Using trans-thoracic echocardiography and HRV recordings, they discovered a significant positive association between ejection fraction and parasympathetic abnormalities. They also found, however, a negative correlation between parasympathetic components and left ventricular volumes. They speculated that autonomic activity might be compensating for the left ventricular dysfunction. Parasympathetic dysfunction, as assessed with spectral analysis, was found to be associated with left ventricular hypertrophy in a study involving 217 diabetic and non-diabetic chronic haemodialysis patients [43^{••}]. Relationships between cardiac ischaemia and autonomic dysfunction have been investigated further by Coquet *et al.* [44], who measured HRV in 32 chronic haemodialysis patients for 24 h and compared this to myocardial scintigraphy markers of ischaemia [44]. Markers of parasympathetic activity were reduced during the non-dialytic period in those that had evidence of myocardial ischaemia, although other differences in other markers of autonomic activity were not in evidence. As already stated, however, markers of autonomic dysfunction generated by spectral analysis do not always remain evident once attempts are made to separate overall function into sympathetic and parasympathetic constituent pathways.

Cardiac arrhythmias, arterial structure and autonomic dysfunction

Sudden cardiac death may be linked to autonomic dysfunction due to a reduction in HRV and intrinsic cardiac responsiveness. Maule *et al.* [45] measured the QT interval from electrocardiograms performed before and after haemodialysis in 69 chronic haemodialysis patients. They concluded that there was considerable QT variability and that the extent of the variability was not related to the presence of autonomic dysfunction. Perhaps utilizing alternative techniques to generate more sensitive information on HRV and cardiac responsiveness may clarify this relationship further. Given that BRS is a potentially remedial factor, it is interesting to note that acetate-free, dual-chamber paired haemodiafiltration, as opposed to

Figure 2. Baroreflex sensitivity and arterial stiffness



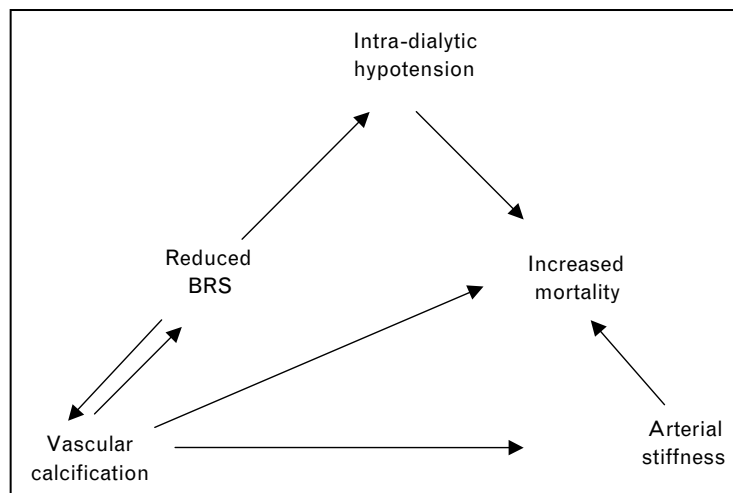
Reduced baroreflex sensitivity (BRS) measurements are associated with shorter time to shoulder (TTS) measurements (representing arterial stiffness), thereby illustrating the links between arterial structure and baroreceptor function. Reproduced with permission [47].

conventional haemodialysis, appears to be associated with a greater BRS response which may contribute to the overall improvement in cardiac stability recognized with this relatively new dialysis technique [46], potentially due to reduced nitric oxide production and direct effects on baroreceptor function.

Vascular calcification is an area of paramount importance within CKD and our group have demonstrated that reduced BRS has been found to be associated with both vascular calcification and arterial stiffness (Fig. 2) [47]. The clinical implication and consequences of autonomic dysfunction are widespread. The baroreceptors involved in the short-term regulation of blood pressure are located in the arterial wall, and thus vascular calcification and arterial stiffness may determine BRS. Impaired sympathovagal balance has been linked to IDH, with previous suggestions that disturbed autonomic control in uraemic patients is linked to cardiac damage [39^{••}]. It is intradialytic haemodynamic instability, (rather than pre-dialysis blood pressure), that has been demonstrated to determine mortality in prospectively studied haemodialysis patients [28^{••}]. Relating arterial structure to function may contribute to the wider understanding of the relationship between vascular calcification, IDH and mortality in haemodialysis patients as suggested in Fig. 3. Covic *et al.* [48^{••}] investigated the relationship between arterial stiffness and endothelial vasomotor function in 40 CAPD, 40 haemodialysis and 20 renal transplant patients. Changes in vasomotor function were assessed by challenges of inhaled salbutamol and sublingual nitroglycerin to represent endothelial-dependent and endothelial-independent changes in arterial stiffness respectively. Perhaps surprisingly, those receiving CAPD

Figure 3. Proposed relationship between arterial structure, function, vascular calcification and increased cardiovascular mortality in chronic haemodialysis patients

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had stiffer arteries than those undergoing haemodialysis and had greater degrees of endothelial-dependent vasomotor function. A prospective study [49] involving 20 patients investigated HRV prior to commencing renal replacement therapy and again 12 months later. Thirteen patients received haemodialysis and seven undertook CAPD. The investigators observed that although HRV prior to commencing renal replacement therapy was reduced in all patients as compared with healthy controls, a significant improvement in time domain analysis parameters was observed in both groups. The increase in this parameter was greater in the group receiving CAPD than in those undergoing chronic haemodialysis. Haemodynamic changes are an underappreciated but characteristic feature of peritoneal dialysis [50], and therefore continuing to investigate cardiovascular control within CAPD as well as haemodialysis remains important.

Conclusion

Vasomotor instability, particularly in those patients receiving chronic haemodialysis, remains an issue of increasing importance. This is particularly pertinent given an ageing and increasingly diabetic dialysis population. Recently, there has been strong evidence that IDH is associated with increased mortality in this population. Autonomic dysfunction is important in the pathogenesis of the complex haemodynamic abnormalities that may exist in CKD patients, including those on peritoneal dialysis. Investigators have recently elucidated associations between reduced HRV, left ventricular dysfunction and hypertrophy, as well as reduced BRS and IDH. Since reduced BRS is potentially amenable to alterations in therapy, further investigation into the significance of autonomic dysfunction in vasomotor instability remains

of prime clinical significance. Relating precise autonomic abnormalities to various patterns of haemodynamic and vasomotor instability has unfortunately not yet been determined and more work is required in this area. Ideally, future studies of both haemodynamic instability and strategies to improve stability should include assessment of autonomic function combined with other, more conventional, cardiovascular markers.

References and recommended reading

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